

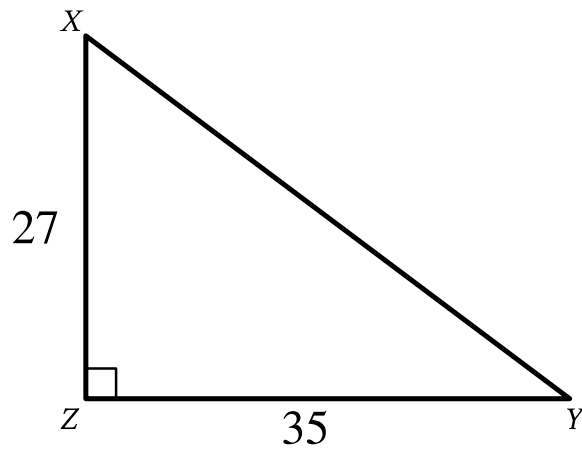
● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

02

10 questions · ~15 min

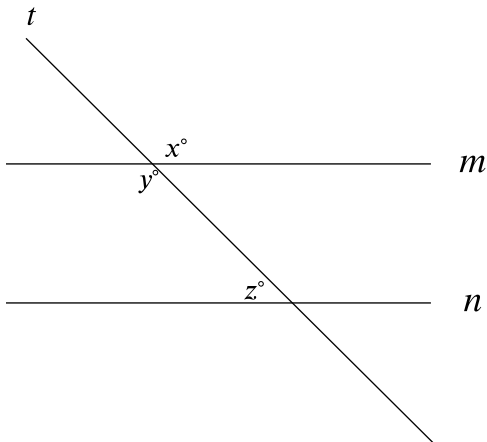


Note: Figure not drawn to scale.

- Angle upper Z is a right angle.
- The length of side upper X upper Z is 27.
- The length of side upper Y upper Z is 35.
- A note indicates the figure is not drawn to scale.

Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

- A. $\tan X$
- B. $\tan Y$
- C. $\cos X$
- D. $\cos Y$



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t , m , and n .
- Line t intersects both line m and line n .
- At the intersection of line t and line m , 2 angles are labeled clockwise from top left as follows:
 - Top right: x°
 - Bottom left: y°
- At the intersection of line t and line n , 1 angle is labeled clockwise from top left as follows:
 - Top left: z°
- A note indicates the figure is not drawn to scale.

In the figure, lines m and n are parallel. If $x = 6k + 13$ and $y = 8k - 29$, what is the value of z ?

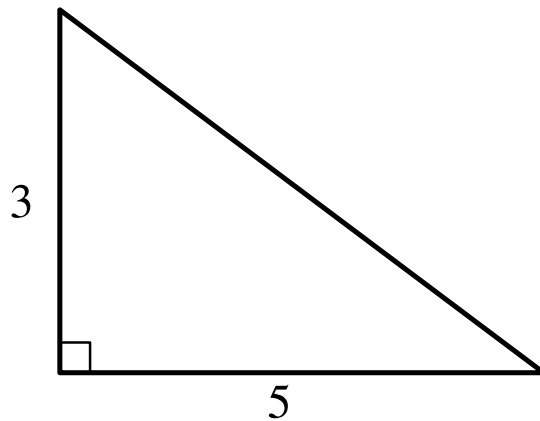
- A. 3
- B. 21
- C. 41

D. 139

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The lengths of the 2 sides adjacent to the right angle are as follows:
 - 3
 - 5
- A note indicates the figure is not drawn to scale.

The figure shows the lengths, in inches, of two sides of a right triangle. What is the area of the triangle, in square inches?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

$\overline{9}$ $\overline{9}$ $\overline{9}$ $\overline{9}$

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN

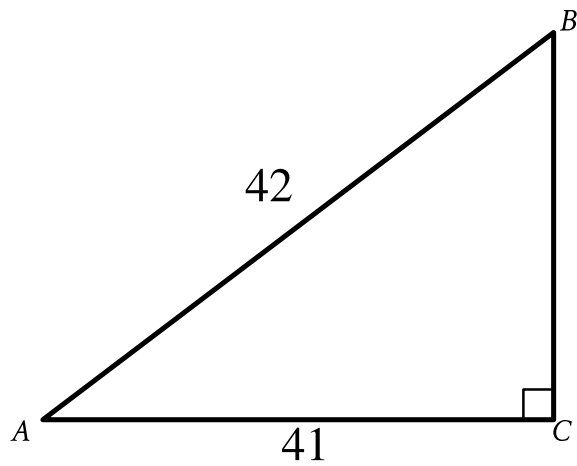
GEOMETRY AND TRIGONOMETRY

In the xy -plane, the graph of the equation $x - 3^2 + y - 5^2 = 9$ is a circle. The point $6, c$, where c is a constant, lies on this circle. What is the value of c ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Angle upper A upper C upper B is a right angle.
- The length of side upper A upper B is 42.
- The length of side upper A upper C is 41.
- A note indicates the figure is not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

- A. $\frac{42}{41}$
- B. $\frac{41}{42}$
- C. $\frac{1}{42}$
- D. $\frac{1}{41}$

In triangle ABC , the measure of angle A is 54° , the measure of angle B is 90° , and the measure of angle C is $\frac{k}{2}^\circ$. What is the value of k ?

- A. 36
- B. 45
- C. 72
- D. 108

A right circular cylinder has a volume of 45π . If the height of the cylinder is 5, what is the radius of the cylinder?

- A. 3
- B. 4.5
- C. 9
- D. 40

The measure of angle R is $\frac{2\pi}{3}$ radians. The measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R . What is the measure of angle T , in degrees?

- A. 75
- B. 120
- C. 195
- D. 390

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin R$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos S$?

- A. $\frac{\sqrt{15}}{15}$
- B. $\frac{\sqrt{15}}{4}$
- C. $\frac{4\sqrt{15}}{15}$
- D. $\sqrt{15}$